Procurement List

Parts still to be purchased for the assembly of the autonomous cargo robot

PROJECT	DOCUMENT No.	REVISION	DATE
AUTONOMOUS CARGO ROBOT – DESIGN v2	**CR-V2-PL-001**	**A**	**12/11/2026**
TITLE	STATUS	SHEET SIZE	UNITS
PROCUREMENT LIST – PARTS TO PURCHASE	**Issued for purchase**	**A4 landscape**	**mm**
INSTITUTION	PREPARED BY	CHECKED BY	APPROVED BY
Arab Academy for Science, Technology & Maritime Transport – Graduation Project II			

1. Scope

This list covers the 44 line items still to be purchased. The 3D-printed parts, the laser-cut plates and the components listed in section 4 are already available and are not included. Part designations and quantities follow the bill of materials of drawing CR-V2-010 and section 0.2 of the build guide.

2. Summary

Sec.	Category	Line items	Remark
1	Drive, main battery and servo supply	3	Order first: longest lead time
2	Battery charging and safety	3	
3	Panel components (mounted through the 6 mm top deck)	3	Must fit a 6 mm panel
4	Power wiring and connectors	6	
5	Signal wiring and electronic parts	7	
6	Fasteners (lengths verified for the 6 mm acrylic plates)	16	Washers on all acrylic
7	Casters, track and consumables	6	
	Total	**44**	

3. General notes

1. All four plates (bottom plate, top deck, tilting plate, cargo plate) are laser-cut from 6 mm acrylic. Fit a washer under every screw head and nut that bears on acrylic, tighten only until snug, and inspect the rod, caster and motor-bracket holes for cracks after the first test runs.
2. The motor-bracket holes are the only holes drilled by hand: use a sharp 3.5 mm bit at low speed with a wooden backing board. Countersink the eight cargo-plate holes slowly.
3. Fastener lengths were checked against the 6 mm plates. The M3 × 12 screws of the line sensor bar and the ultrasonic bracket were too short and are replaced by M3 × 16; the servo-horn screws are M3 × 8 (not M3 × 10).
4. Δ in the Rev. column marks an item changed for the 6 mm acrylic plates (compared with the issue of 10/11/2026). The build guide and the drawing set (CR-V2-001 to CR-V2-010) are revised to match.
5. Quantities include spares only where stated. Overall height with the bed flat: 220 mm.

Parts to purchase

Rcvd	Item	Description	Specification	Qty	Unit	Application	Rev.
1 DRIVE, MAIN BATTERY AND SERVO SUPPLY							
☐	1.1	**JGB37-520 gear motor set, 12 V, 110 RPM**	Motor with Hall encoder (6-wire cable), steel L-bracket, hex coupling and Ø65 mm wheel. Output shaft Ø6 mm, D-profile (often listed as "5.5 mm D-shaft"). Both sets identical.	2	set	Drive, left and right. Two driven wheels; the two ball casters carry the rest.	
☐	1.2	**LiPo battery, 3S**	11.1 V, 5000 mAh, discharge rating ≥ 20 C, XT60 connector, with balance lead.	1	pc	Main power supply. A second battery is recommended for testing.	
☐	1.3	**Buck converter XL4016**	8 A, adjustable output, with heat sink. Set to 7.4 V before connecting the servos (max. 8.4 V).	1	pc	Servo supply, kept separate from the 5 V logic supply.	
2 BATTERY CHARGING AND SAFETY							
☐	2.1	**LiPo balance charger**	IMAX B6AC class (mains supply built in). The plain B6 also needs a 12 V, ≥ 5 A DC adapter.	1	pc	Charging the 3S battery.	
☐	2.2	**LiPo safety bag**	Fire-resistant.	1	pc	Charging and storing the battery.	
☐	2.3	**LiPo voltage alarm**	1–8 S, buzzer, connects to the balance lead. Set to 3.5 V per cell.	1	pc	Low-battery warning.	
3 PANEL COMPONENTS (MOUNTED THROUGH THE 6 MM TOP DECK)							
☐	3.1	**Emergency-stop push button**	Ø22 mm panel mount, mushroom head, latching, normally-closed (NC) contact block, 10 A. Must clamp a 6 mm panel.	1	pc	Cuts motor and servo power.	**Δ**
☐	3.2	**Toggle switch**	12 mm panel mount, rated ≥ 15 A at 12 V DC. Thread length ≥ 9 mm.	1	pc	Master power switch.	**Δ**
☐	3.3	**Active buzzer module**	3-pin, 3.3–5 V.	1	pc	Audible signals.	
4 POWER WIRING AND CONNECTORS							
☐	4.1	**Inline blade-fuse holder with fuses**	Holder for 14 AWG wire; 10 A blade fuses.	1 + 5	pc	Battery protection.	
☐	4.2	**XT60 connector pair**	Male and female.	4	pair	Battery and power connections.	
☐	4.3	**WAGO 221-415 lever connector**	5-way.	8	pc	Power distribution buses.	
☐	4.4	**WAGO 221-413 lever connector**	3-way.	6	pc	Power distribution buses.	
☐	4.5	**Silicone wire, 16 AWG**	Red and black.	3 + 3	m	Battery, fuse, switch, E-stop, motor drivers, servo supply.	
☐	4.6	**Silicone wire, 20 AWG**	Red and black.	2 + 2	m	Buck converters, motors.	
5 SIGNAL WIRING AND ELECTRONIC PARTS							
☐	5.1	**Jumper wires**	20 cm, female–female and male–female.	40 + 40	pc	Signal connections.	
☐	5.2	**Servo extension lead**	30 cm.	2	pc	Servos to the I/O board.	

Rcvd	Item	Description	Specification	Qty	Unit	Application	Rev.
☐	5.3	**USB cable**	Data-capable (not charge-only), ≥ 1 m, connector to suit the ESP32 board.	1	pc	Programming the ESP32.	
☐	5.4	**Perfboard**	50 × 70 mm.	2	pc	I/O board (1 spare).	
☐	5.5	**Wire ferrule kit with crimping tool**	Assorted sizes for 16–22 AWG.	1	set	Wire ends in screw terminals.	
☐	5.6	**Cable management kit**	Heat-shrink assortment, cable ties, spiral wrap, 20 mm hook-and-loop strap (1 m), double-sided foam tape.	1	set	Wiring; battery retention.	
☐	5.7	**VHB mounting tape**	19 mm wide.	1	roll	Mounting the small boards without drilling.	
6 FASTENERS (LENGTHS VERIFIED FOR THE 6 MM ACRYLIC PLATES)							
☐	6.1	**M6 threaded rod**	Zinc-plated steel, 1 m. Cut into 6 pieces of 130 mm.	1	pc	Frame columns.	
☐	6.2	**M6 hex nut**	Zinc-plated steel.	26	pc	Frame columns (24 + 2 spare).	
☐	6.3	**M6 washer**	Zinc-plated steel.	26	pc	Frame columns (24 + 2 spare).	
☐	6.4	**M8 × 50 bolt with 2 nuts and 2 washers**	Zinc-plated steel.	2	set	Ball casters.	
☐	6.5	**M4 × 10 screw**		2	pc	Bed rests.	
☐	6.6	**M4 × 12 screw**	Fitted without a washer (needs its full length to engage the nut).	6	pc	Servo mounts.	**Δ**
☐	6.7	**M4 × 16 screw**		4	pc	Servo ear clamps.	
☐	6.8	**M4 × 20 screw**		12	pc	Horn adapters (4), bed risers from below (8).	
☐	6.9	**M4 × 20 countersunk screw**	Flat head, flush with the cargo plate surface.	8	pc	Cargo plate to bed risers.	
☐	6.10	**M4 hex nut**		32	pc	All M4 joints.	
☐	6.11	**M4 washer**		32	pc	All M4 joints.	
☐	6.12	**M3 × 8 screw**	Maximum length 8 mm (a longer screw hits the adapter flange).	8	pc	Disc horns to horn adapters.	
☐	6.13	**M3 × 8 countersunk screw**		6	pc	Caster rings to caster cups.	
☐	6.14	**M3 × 12 screw with nut**		11	set	Motor brackets (8), IR modules (3).	
☐	6.15	**M3 × 16 screw with nut**	Replaces M3 × 12 here (too short through the part, plate and nut).	4	set	Line sensor bar (2), ultrasonic bracket (2).	**Δ**
☐	6.16	**M3 washer**		30	pc	M3 joints on the acrylic bottom plate.	**Δ**
7 CASTERS, TRACK AND CONSUMABLES							
☐	7.1	**Steel ball**	Ø25.4 mm (1 in), bearing grade. A Ø25 mm glass marble is acceptable.	3	pc	Ball casters (2 + 1 spare).	
☐	7.2	**Floor tape, matte black**	19 mm wide (50 mm is too wide: it reads as a station mark).	1	roll	Line-following track.	
☐	7.3	**Masking tape**		1	roll	Track layout.	

Rcvd	Item	Description	Specification	Qty	Unit	Application	Rev.
☐	7.4	**Rubber pad, self-adhesive**	Ø18 mm × 2 mm.	2	pc	Bed rests.	
☐	7.5	**Thread-locker, medium strength**	Blue (removable).	1	pc	Coupling set screws, horn screws.	
☐	7.6	**Grease**	Petroleum jelly is suitable.	1	pc	Caster cups.	

Rcvd = tick when received. Δ = changed for the 6 mm acrylic plates. pc = piece.

4. Components in stock (not included in this list)

No.	Component	Remarks
1	**3D-printed parts (PETG)**	Complete set: servo mounts, clamps, horn adapters, bed rests, risers, line sensor bar, ultrasonic bracket, caster cups and rings.
2	**Laser-cut plates, 6 mm acrylic**	Bottom plate, top deck, tilting plate, cargo plate.
3	**ESP32 DevKitC 38-pin**	2 pcs (1 spare), with the 38-pin screw-terminal breakout board.
4	**BTS7960 (IBT-2) motor driver**	2 pcs.
5	**XL4015 5 A buck converter**	1 pc. Logic supply, set to 5.1 V. No spare (by decision).
6	**DS5160 servo, 60 kg·cm**	2 pcs, with round aluminium disc horns (fit the printed horn adapters).
7	**Sensors**	3-pin IR line sensor modules and the HC-SR04 ultrasonic sensor.
8	**Start push button and status LEDs**	12 mm momentary push button (normally open); 5 mm LEDs, green and red.
9	**Resistors and capacitor**	For the I/O board. Required: 1 kΩ × 3, 2 kΩ × 3, 10 kΩ × 5, 2.2 kΩ × 1, 330 Ω × 2 (¼ W) and one 100 nF ceramic capacitor.
10	**Tools**	Soldering iron, multimeter, drill and bits, spanners.
11	**Test track floor**	Light-coloured floor for the black line.

5. Open items

No.	Subject	Action	Status
1	**Start button and LEDs in stock (section 4)**	Check that the push button's thread is at least 9 mm long, so that it passes through the 6 mm deck and still takes its nut, and that the LED clips fit a 6 mm panel (otherwise bond the LEDs into their holes with hot glue).	Open

6. Approval

	Name	Signature	Date
Prepared by			
Checked by			
Approved by			